

Architecture Tradeoffs and Tech Debt

Confirmed compromises, coupling, risks, removals, and remediation guidance.

Eurskem AI · Agentic Workflow Platform

Generated 2026-08-24

Code revision 8bd16ca4a2a3

Derived from the live repository

Source of truth. Inventories and symbol tables are generated from the current checkout. Narrative explains observed structures and does not replace executable contracts.

Confirmed tradeoffs

| Decision | Benefit | Cost / debt | |---|---|---| | Shared services mapping for nodes | Easy construction and test fakes | String-key contracts and runtime missing-service checks | | MongoDB document stores | Flexible evolving run/workflow records | Cross-document invariants require explicit indexes and code | | Redis for coordination concerns | One dependency for leases, streams, limits, checkpoints | Redis outage has broad operational impact | | Local React state/ effects | Low framework overhead | Repeated loading/error patterns and possible stale state | | Schema-driven node registry | Builder/runtime alignment | Registry imports and manifest generation are critical paths | | Generated Chat adapters | Reuses governed workflow runtime | Cache identity must include every semantic scope |

Current debt and risk

- Some older prose predates removal of Pipeline and Business View subsystems; generated inventories are authoritative.
- Alert routing is less complete than alert rule generation.
- Application orchestration spans routes, runtime helpers, and stores rather than one service layer.
- Large/high-branch modules should only be decomposed behind regression coverage.
- Generated QA artifacts need an explicit retention policy.

Symbol	Kind	Location	Signature / summary
<code>_HorizonDocxBUILDER</code>	class	<code>app/tools/docx_proposal_rendering.py:860</code>	 Provides the HorizonDocxBUILDER behaviour.
<code>ProposalEvidenceFactoryAgent</code>	class	<code>app/nodes/proposal_evidence_factory.py:351</code>	<code>NodeType</code> Workflow node type implementing the ProposalEvidenceFactoryAgent capability.
<code>RegistryLLMGateway</code>	class	<code>app/llm/registry.py:622</code>	<code>LLMGateway</code> LLMGateway adapter that dispatches every call through the registry.
<code>plan_workspace</code>	function	<code>app/workflow/chat_workspace_planner.py:773</code>	<pre>objective: str, *, experience_id: str None=None, skill_name: str None=None, selected_workflow: str None=None, preferred_output: OutputKind='auto', has_attachments: bool=False, attachment_categories: list[str] None=None, collection_id: str None=None, retrieval_profile_id: str None=None, rag_agent_id: str None=None, document_ids: list[str] None=None, integration_connection: str None=None, integration_tool: str None=None, previous_run_id: str None=None, previous_result: Any=None</pre> No summary declared.
method			

Symbol	Kind	Location	Signature / summary
ProposalEvidenceFactoryAgent.run		app/nodes/proposal_evidence_factory.py:374	<code>self, state: dict[str, Any], resolved_config: dict[str, Any]</code> Run the result.
RouterAgent	class	app/nodes/router.py:254	<code>NodeType</code> Workflow node type implementing the RouterAgent capability.
RetrievalService	class	app/retrieval/service.py:69	<code></code> Provides the RetrievalService behaviour.
lifespan	async function	app/main.py:104	<code>app: FastAPI</code> Compute the lifespan.
TransformAgent	class	app/nodes/transform.py:301	<code>NodeType</code> Workflow node type implementing the TransformAgent capability.
IngestionJobRunner	class	app/ingestion/jobs.py:36	<code></code> Runs without a second queue platform; progress lives in Mongo.
WorkflowSpec	class	app/runtime/schema.py:383	<code>BaseModel</code> The stable, use-case-neutral workflow contract.
WebSearchService	class	app/tools/web_io.py:65	<code></code> Provides the WebSearchService behaviour.
RAGService	class	app/rag/service.py:75	<code></code> Provides the RAGService behaviour.
RetrievalService._retrieve_impl	method	app/retrieval/service.py:211	<code>self, request: RetrievalQuery, *, owner_scope_id: str, request_id: str, llm: Any None=None</code> Retrieve the impl.
inspect_python	function	scripts/build_documentation_pdf.py:234	<code>path: Path, model: Model</code> No summary declared.
_probe_services	async function	app/runtime/preflight.py:2356	<code>spec: WorkflowSpec, services: dict[str, Any], report: WorkflowPreflightReport, *, require_run_history: bool, owner_scope_id: str None=None</code> Probe the services.
BoundedToolResearchService	class	app/research/deep_research.py:199	<code></code> Bounded, in-process replacement for the OpenAI Deep Research job API.
_make_runtime_fn	function	app/runtime/compiler.py:63	<code>instance, bus: RunEventBus None, services: dict</code> Wrap a NodeType instance so it conforms to LangGraph's 'state -> partial state' contract.
MCPIntegrationService	class	app/mcp/service.py:99	<code></code> Provides the MCPIntegrationService behaviour.
_wire_edges	function	app/runtime/compiler.py:428	<code>graph: StateGraph, edges: list[EdgeSpec], hitl_ids: set[str], multi_router_ids: set[str] None=None</code> Add edges, return the set of source node ids (used to compute terminals).
IngestionJobRunner.run	method	app/ingestion/jobs.py:115	<code>self, job: IngestionJob, paths: list[Path], *, metadata: dict[str, Any]</code>

Symbol	Kind	Location	Signature / summary
			None=None Run the result.
RunEventBus	class	app/runtime/events.py:82	Provides the RunEventBus behaviour.
DOCXProposalRenderer	class	app/nodes/docx_renderer.py:120	NodeType Workflow node type implementing the DOCXProposalRenderer capability.
_apply	function	app/nodes/data_transform.py:299	op: TransformOperation, context: dict[str, Any] Apply the result.
_validate_template_output_path	function	app/runtime/preflight.py:1635	spec: WorkflowSpec, current_node: NodeSpec, reference: str, report: WorkflowPreflightReport, path: str, forward: dict[str, set[str]], guaranteed: dict[str, set[str]], optional: bool=False Validate the template output path.
WorkflowSpec.validate_graph_references	method	app/runtime/schema.py:406	self Validate the graph references.
RAGService.query	method	app/rag/service.py:147	self, *, owner_scope_id: str, rag_agent_id: str, query: str, runtime_filters: dict[str, Any] MetadataFilterGroup None=None, document_ids: list[str] None=None, runtime_context: dict[str, Any] None=None, llm: Any None=None Query the result.
BoundedToolResearchService.research	method	app/research/deep_research.py:241	self, *, brief: ResearchBrief, instructions: str, max_duration_seconds: float=1800.0, max_iterations: int=15, max_cost_per_call_usd: float=15.0 Compute the research.
OpenAllImageGenerationService	class	app/tools/image_io.py:64	Provides the OpenAllImageGenerationService behaviour.
validate_field_constraints	function	app/nodes/workflow_input.py:454	bindings: list[InputFieldBinding], data: dict[str, Any] Business-friendly validation beyond shape (§20/21 — never a raw Pydantic path like VALIDATION_SCHEMA_FIELD_002): text length/pattern, email/url/phone format, percentage range, and the semantic checks a compound preset's own object shape can't express on its own (date_range ordering, an allowed unit/
Settings	class	app/config.py:15	BaseSettings Provides the Settings behaviour.
_ScholarlyCandidateDiscovery	class	app/nodes/evidence_agent.py:161	NodeType Workflow node type implementing the ScholarlyCandidateDiscovery capability.
extract_identifiers	function	app/evidence/identifiers.py:173	url: str None=None, *, metadata: dict[str, Any] None=None, text: str None=None Best-effort offline identifier extraction.

Symbol	Kind	Location	Signature / summary
<code>_check</code>	function	<code>app/nodes/consistency_checker.py:57</code>	<code>graph: ProposalGraph</code> Check the result.
<code>EntityTokenizerService</code>	class	<code>app/security/entity_tokenizer.py:137</code>	 Provides the EntityTokenizerService behaviour.
<code>analyze_chat_output</code>	function	<code>app/workflow/chat_output_compatibility.py:40</code>	<code>spec: WorkflowSpec</code> Describe whether a workflow has a meaningful normalized Chat result.
<code>Settings.validate_production_security</code>	method	<code>app/config.py:442</code>	<code>self</code> Refuse to boot production with development security settings.
<code>IntegrationService</code>	class	<code>app/integrations/files/service.py:105</code>	 One capability, many providers, one place that owns token refresh.
<code>ScientificSkillCatalog</code>	class	<code>app/research/skills.py:215</code>	 Read and select only configured, locally installed skill documents.
<code>start_ingestion</code>	async function	<code>app/api/knowledge.py:386</code>	<code>collection_id: str, request: Request, files: list[UploadFile]=File(...), parser_profile_id: str None=Form(None), parser_profile_version: int None=Form(None), chunking_profile_id: str None=Form(None), chunking_profile_version: int None=Form(None), embedding_profile_id: str None=Form(None), embedding_profile_version: int None=Form(None), embedding_model: str None=Form(None), metadata_json: str=Form('{}'), user: CurrentUser=Depends(require_permission('knowledge:write'))</code> Start the ingestion.
<code>build_call_coverage_matrix</code>	function	<code>app/proposal_graph/coverage.py:88</code>	<code>graph: ProposalGraph</code> Build the call coverage matrix.
<code>_select</code>	function	<code>scripts/openrouter_health_check.py:667</code>	<code>catalog: list[CatalogEntry], args: argparse.Namespace</code> Returns (selected, excluded) — excluded rows are reported, never silently dropped.
<code>KnowledgeRepository</code>	class	<code>app/knowledge/repository.py:75</code>	 One instance shared across the app via <code>services["knowledge_repository"]</code> .
<code>DataverseClient</code>	class	<code>app/mcp/dynamics/client.py:189</code>	<code>DynamicsBackend</code> Live Dynamics 365 over the Dataverse Web API.
<code>MCPToolAgent</code>	class	<code>app/nodes/mcp_tool.py:154</code>	<code>NodeType</code> Workflow node type implementing the MCPToolAgent capability.
<code>evaluate_condition</code>	function	<code>app/runtime/rules.py:634</code>	<code>condition: Condition, context: dict[str, Any]</code> Compute the evaluate condition.
<code>OpenAIGateway</code>	class	<code>app/llm/openai_gw.py:165</code>	<code>LLMGateway</code> OpenAI-backed gateway.

Symbol	Kind	Location	Signature / summary
<code>_explain</code>	function	<code>app/api/builder.py:739</code>	<code>type_name: str, output: dict[str, Any]</code> Turn a node's output into the "why did this happen?" view (§24, §47).
<code>CitationRegistryBuilder</code>	class	<code>app/nodes/citation_registry_builder.py:134</code>	<code>NodeType</code> Workflow node type implementing the CitationRegistryBuilder capability.
<code>CitationRegistryBuilder.run</code>	method	<code>app/nodes/citation_registry_builder.py:152</code>	<code>self, state: dict[str, Any], resolved_config: dict[str, Any]</code> Run the result.
<code>EmailService</code>	class	<code>app/integrations/email/service.py:100</code>	 One capability, many providers, one place that owns the safety rules.
<code>LocalOpenAICompatibleGateway</code>	class	<code>app/llm/local_openai_gw.py:110</code>	<code>LLMGateway</code> Gateway used by one configured local/private model endpoint.
<code>OpenRouterGateway</code>	class	<code>app/llm/openrouter_gw.py:124</code>	<code>LLMGateway</code> OpenRouter-backed gateway.
<code>build_rows</code>	function	<code>scripts/build_workflow_qa_matrix.py:127</code>	<code>root: Path, preflight_reports: dict[str, dict[str, Any]]</code> No summary declared.
<code>run</code>	function	<code>scripts/verify_knowledge_studio.py:474</code>	<code>base: str, user: str, password: str, keep: bool</code> Run the result.
<code>CostLedger</code>	class	<code>app/observability/cost_ledger.py:161</code>	 Provides the CostLedger behaviour.
<code>_validate_graph</code>	function	<code>app/runtime/preflight.py:1460</code>	<code>spec: WorkflowSpec, report: WorkflowPreflightReport</code> Validate the graph.
<code>PublicDatabaseLookupService</code>	class	<code>app/tools/database_lookup.py:42</code>	 Rate-limited, allow-listed public database transport.
<code>MicrosoftGraphAdapter</code>	class	<code>app/integrations/email/msgraph.py:41</code>	<code>EmailAdapter</code> Provides the MicrosoftGraphAdapter behaviour.
<code>_ScholarlyCandidateDiscovery.run</code>	method	<code>app/nodes/evidence_agent.py:179</code>	<code>self, state: dict[str, Any], resolved_config: dict[str, Any]</code> Run the result.
<code>InternalProjectEvidenceRetrieverAgent</code>	class	<code>app/nodes/internal_project_evidence_retriever.py:126</code>	<code>NodeType</code> Workflow node type implementing the InternalProjectEvidenceRetrieverAgent capability.
<code>KnowledgeRetrieval</code>	class	<code>app/nodes/knowledge_retrieval.py:90</code>	<code>NodeType</code> Workflow node type implementing the KnowledgeRetrieval capability.
<code>_chunk_from_object</code>	function	<code>app/retrieval/strategies.py:16</code>	<code>obj: Any, *, score_kind: str, rank: int</code> Chunk the from object.
<code>_register_builders</code>	function	<code>app/runtime/logic_preflight.py:52</code>	 Populate the typed-output builders lazily.
<code>_validate_router_edges</code>	function	<code>app/runtime/preflight.py:828</code>	<code>spec: WorkflowSpec, report: WorkflowPreflightReport</code> Validate the router edges.

Symbol	Kind	Location	Signature / summary
PdfPageVisionDescriber	class	app/ingestion/vision.py:56	Render PDF pages and describe their visual content with a vision model.
BoundedDeepResearchAgent	class	app/nodes/bounded_deep_research_agent.py:117	NodeType Workflow node type implementing the BoundedDeepResearchAgent capability.
InternalProjectEvidenceRetrieverAgent.run	method	app/nodes/internal_project_evidence_retriever.py:153	self, state: dict[str, Any], resolved_config: dict[str, Any] Run the result.
RAGAgent	class	app/nodes/rag.py:156	NodeType Workflow node type implementing the RAGAgent capability.
StructuredDatasetRetrieverAgent	class	app/nodes/structured_dataset_retriever.py:139	NodeType Workflow node type implementing the StructuredDatasetRetrieverAgent capability.
resume_workflow_durable	async function	app/runtime/hitl.py:109	run_id: str, decision: dict[str, Any], *, services: dict[str, Any] None=None, session_id: str None=None, actor: str='system' Resume in memory when possible, otherwise rebuild from Mongo.
_validate_start_fields	function	app/runtime/preflight.py:1365	spec: WorkflowSpec, report: WorkflowPreflightReport Validates one Start node's own field/file_field list.
_HorizonDocxBuilder._append_inline	method	app/tools/docx_proposal_rendering.py:1413	self, paragraph, container, style: InlineStyle None=None Append the inline.
_adjacency	function	app/nodes/about_synthesis.py:77	type_name -> {"upstream": [...], "downstream": [...], "example": str None} , mined once from every workflow YAML on disk.
AITaskAgent	class	app/nodes/ai_task.py:424	NodeType Workflow node type implementing the AITaskAgent capability.
PriorProjectRetrieverAgent	class	app/nodes/prior_project_retriever.py:139	NodeType Workflow node type implementing the PriorProjectRetrieverAgent capability.
PriorProjectRetrieverAgent.run	method	app/nodes/prior_project_retriever.py:163	self, state: dict[str, Any], resolved_config: dict[str, Any] Run the result.
ProposalSubmissionGate	class	app/nodes/proposal_submission_gate.py:108	NodeType Combine hard proposal checks into one auditable release decision.
ProposalSubmissionGate.run	method	app/nodes/proposal_submission_gate.py:125	self, state: dict[str, Any], resolved_config: dict[str, Any] Run the result.
RAGAgent.run	method	app/nodes/rag.py:178	self, state, resolved_config: dict[str, Any] Run the result.
WorkflowBuilderStore	class		

Symbol	Kind	Location	Signature / summary
		app/workflow/ builder_store.py:60	Own autosave drafts and immutable versions for one workflows root.
AnthropicGateway	class	app/llm/anthropic_gw.py:114	LLMGateway Live Anthropic Claude gateway.
MinIOEvidenceIngestion	class	app/nodes/ minio_evidence_ingestion.py:108	NodeType Workflow node type implementing the MinIOEvidenceIngestion capability.
MinIOEvidenceIngestion.run	method	app/nodes/ minio_evidence_ingestion.py:126	self, state: dict[str, Any], resolved_config: dict[str, Any] Run the result.
ProposalEvidenceFactoryAgent.quality_report	method	app/nodes/ proposal_evidence_factory.py:1235	*, policy: EvidencePolicy, verified_claims: list[VerifiedClaim], citation_registry: list[CitationRegistryEntry], links: list[ClaimEvidenceLink], audit: list[SearchAuditRecord], rejected: list[RejectedCandidate], quantitative_registry: list[dict[str, Any]] Internal helper for the quality report step.
ProposalWorkspaceStore	class	app/proposal_graph/ workspace_store.py:129	Provides the ProposalWorkspaceStore behaviour.
_check_conditional_join_reachability	function	app/runtime/ logic_preflight.py:873	spec: WorkflowSpec, issue: Callable[..., None] Two conditional dispatches into one shared step form an unsatisfiable join.
_HorizonDocxBuilder.render_block	method	app/tools/ docx_proposal_rendering.py:1251	self, node, *, list_level: int=0 Render the block.
OpenAllImageGenerationService.generate_openrouter	method	app/tools/image_io.py:178	self, prompt: str, *, size: str, quality: str, output_format: str, model: str No summary declared.
backfill	async function	scripts/ backfill_knowledge_resources.py:85	*, apply: bool, db=None Backfill the result.
cost_overview	async function	app/api/cost_admin.py:531	request: Request, days: int=Query(default=30, ge=1, le=365), user: CurrentUser=Depends(require_admin) Compute the cost overview.
ask_about_node_types	async function	app/api/ node_types_chat.py:153	req: AskAboutNodeTypesRequest, request: Request, user: CurrentUser=Depends(require_consultant) Compute the ask about node types.
_build_run_context	function	app/api/run_chat.py:116	run: dict[str, Any] Build the run context.
InMemoryEmailAdapter	class	app/integrations/email/ memory.py:29	EmailAdapter Provides the InMemoryEmailAdapter behaviour.
ExternalActionService	class		

Symbol	Kind	Location	Signature / summary
		app/integrations/ external_action.py:62	Provides the ExternalActionService behaviour.
StructuredDatasetRetrievalAgent.run	method	app/nodes/ structured_dataset_retrieval.py:166	self, state: dict[str, Any], resolved_config: dict[str, Any] Run the result.
transform_query	async function	app/retrieval/ query_transform.py:55	request: RetrievalQuery, *, llm: Any, metadata_schema: dict[str, Any] Return "(semantic_query, transformed_queries, generated_filters)".
FieldSpec	class	app/runtime/ field_schema.py:67	BaseModel One row in the visual schema builder.
_validate_nodes	function	app/runtime/ preflight.py:488	spec: WorkflowSpec, report: WorkflowPreflightReport Validate the nodes.
_probe_workflow_model_access	async function	app/runtime/ preflight.py:2860	spec: WorkflowSpec, llm: Any, report: WorkflowPreflightReport Confirm provider/project access without making a generation request.
PublicDatabaseLookupService.query_eurostat	method	app/tools/ database_lookup.py:72	self, query: StructuredDatasetQuery, *, timeout_seconds: float=45.0, max_response_bytes: int=20 * 1024 * 1024 Query the eurostat.
retry_failed_run	async function	app/api/runs.py:253	source_run_id: str, req: RetryRunRequest, request: Request, user: CurrentUser=Depends(require_consultant) Create a new attempt that reuses every completed node checkpoint.
BoundedDeepResearchAgent.run	method	app/nodes/ bounded_deep_research_agent.py:141	self, state: dict[str, Any], resolved_config: dict[str, Any] Run the result.
ClaimEvidenceVerifier	class	app/nodes/ claim_evidence_verifier.py:62	NodeType Workflow node type implementing the ClaimEvidenceVerifier capability.
SubprocessAgent	class	app/nodes/subprocess.py:115	NodeType Workflow node type implementing the SubprocessAgent capability.
run_workflow	async function	app/runtime/executor.py:83	spec: WorkflowSpec, inputs: dict[str, Any], session_id: str None=None, collection_id: str='default', services: dict[str, Any] None=None, run_id: str None=None, reused_node_results: dict[str, dict[str, Any]] None=None, retry_source_run_id: str None=None, hit_resume_decisions: dict[str, dict[str, Any]] None=None, resume_replay: bool=False Run the workflow.
_node_autofix_scope_valid	function	app/api/workflows.py:358	original_yaml: str, candidate_yaml: str, node_id: str Only the selected node's configuration/model controls may change.

Symbol	Kind	Location	Signature / summary
IntegrationService.execute	method	app/integrations/files/service.py:262	<pre>self, *, connection_id: str, operation: IntegrationOperation, folder_id: str list[str] None=None, file_id: str list[str] None=None, query: str='', page_size: int=25, page_token: str None=None</pre> Execute the result.
validate_external_action_url	function	app/integrations/url_guard.py:42	<pre>url: str, *, allow_http: bool=False, resolve: Callable[... , list]=socket.getaddrinfo</pre> Validate an outbound External Action URL and return it unchanged.
HumanInLoopAgent	class	app/nodes/human_in_loop.py:157	NodeType Workflow node type implementing the HumanInLoopAgent capability.
IntegrationAgent	class	app/nodes/integration.py:206	NodeType Workflow node type implementing the IntegrationAgent capability.
_decode_jsonstat	function	app/nodes/structured_dataset_retriever.py:436	<pre>*, query: StructuredDatasetQuery, payload: dict[str, Any], endpoint: str, parameters: dict[str, Any], accessed_at: str, response_sha256: str, snapshot_object_key: str, limit: int, source_version: str</pre> Decode the jsonstat.
_check_subprocess_agents	function	app/runtime/logic_preflight.py:1642	<pre>spec: WorkflowSpec, issue: Callable[... , None], guaranteed_before: Callable[[str, str], bool] None</pre> Everything about a Subprocess step preflight can prove without actually running the child: the referenced workflow exists and parses, every explicit input mapping targets a real declared input of it, every *required* input of it resolves to something, and the whole reference graph this workflow reacts to.
_validate_multiroute_andjoin	function	app/runtime/preflight.py:1238	<pre>spec: WorkflowSpec, report: WorkflowPreflightReport, forward: dict[str, set[str]]</pre> A fan-in target whose declared predecessors depend on DIFFERENT Multi-Route selections can silently never fire, the same failure shape FANIN_UNREACHABLE_ANDJOIN already catches for ordinary (single-select) routers' mutually-exclusive branches — confirmed empirically against real langgraph 1.2.9: a target dispatched to only a SUBSET of a multi-router's own branches is never revisited for the branches that weren't selected, so any node whose AND-join lists all of them hangs forever on the ones that never ran.
probe_model	async function	scripts/openrouter_health_check.py:530	<pre>gateway: OpenRouterGateway, entry: CatalogEntry, *, probes: tuple[str, ...], timeout: float, retry_delay: float</pre> Probe the model.
run	async function	scripts/openrouter_health_check.py:1034	<pre>args: argparse.Namespace</pre> Run the result.

Symbol	Kind	Location	Signature / summary
restart_run	async function	app/api/runs.py:580	<code>source_run_id: str, req: RetryRunRequest, request: Request, user: CurrentUser=Depends(require_consultant)</code> Create a brand-new run of the same workflow and original inputs.
_field_type_label	function	app/api/workflow_generation.py:410	<code>schema: dict</code> A short type descriptor for one JSON-schema property, e.g.
KnowledgeService	class	app/knowledge/service.py:83	 Provides the KnowledgeService behaviour.
MCPClient	class	app/mcp/client.py:224	 Multi-server MCP wrapper.
_fixture_matches	function	app/mcp/dynamics/client.py:716	<code>row: dict[str, Any], expression: str</code> Interpret the small OData subset this package generates.
HumanInLoopAgent.run	method	app/nodes/human_in_loop.py:211	<code>self, state, resolved_config: dict[str, Any]</code> Run the result.
ScientificResearchPlanner Agent	class	app/nodes/scientific_research_planner.py:134	<code>NodeType</code> Workflow node type implementing the ScientificResearchPlannerAgent capability.
ScientificResearchPlanner Agent.run	method	app/nodes/scientific_research_planner.py:157	<code>self, state: dict[str, Any], resolved_config: dict[str, Any]</code> Run the result.
ObjectStore	class	app/storage/minio_client.py:75	 S3-compatible object store wrapper.
workflow_stats	async function	app/workflow/run_history.py:1246	<code>db, session_id: str, workflow_name: str, *, sample_limit: int=200</code> Library "Runs and performance" tab: success rate, median duration, and the most common failure, computed from this session's own run history for one workflow.
_resolve_pending_gate	async function	app/api/runs.py:123	<code>db: Any, scope: str, run_id: str, *, parent_run_id: str, visited: set[str] None=None</code> Follow subprocess waits to the deepest actionable human gate.
_node_type_catalog	function	app/api/workflow_generation.py:446	<code>type_names: list[str] None=None</code> The catalog text embedded in the generation system prompt.
candidate_from_paper	function	app/evidence/retrieval.py:202	<code>paper: dict[str, Any], *, claim_id: str, query: str, purpose: str, source_hint: str None=None, discovery_lane: str None=None</code> Compute the candidate from paper.
deduplicate_candidates	function	app/evidence/retrieval.py:373	<code>candidates: list[CandidateSource]</code> Deduplicate the same work for one claim, across queries AND lanes.
DelimitedTextExtractor	class	app/ingestion/extractor.py:393	 Extract rows from a CSV/TSV file, one ExtractedUnit per row block.

Symbol	Kind	Location	Signature / summary
DelimitedTextExtractor.extract	method	app/ingestion/extractor.py:414	<code>self, path: Path</code> Extract the result.
ingest_file	async function	app/ingestion/pipeline.py:59	<code>path: Path, metadata: dict[str, Any] None=None, chunk_config: ChunkConfig None=None, object_store: ObjectStore None=None, embedder: Embedder None=None, weaviate: WeaviateClient None=None, mongo: MongoClient None=None, collection_config: CollectionConfig None=None</code> Ingest one document end-to-end.
ModelRouter	class	app/llm/model_router.py:175	<code></code> Choose the most suitable configured model without spending tokens.
MCPToolAgent._required_arguments	method	app/nodes/mcp_tool.py:326	<code>self, service: Any, cfg: MCPToolConfig</code> The argument names the server itself declares as required.
StartAgent	class	app/nodes/start.py:117	<code>NodeType</code> Workflow node type implementing the StartAgent capability.
ScientificSkillCatalog.select	method	app/research/skills.py:314	<code>self, *, objective: str, requested: Iterable[str]=(), auto_select: bool=True, max_skills: int=3</code> Select the result.
build_field_index	function	app/runtime/logic_preflight.py:743	<code>spec: WorkflowSpec</code> Typed paths for every node in the workflow.
_validate_inputs	function	app/runtime/preflight.py:2014	<code>spec: WorkflowSpec, provided_inputs: dict[str, Any] None, report: WorkflowPreflightReport</code> Validate the inputs.
EntityTokenizerService.detect_spans	method	app/security/entity_tokenizer.py:211	<code>self, text: str, *, session_id: str, collection_id: str</code> Detect the spans.